

Block I

. 1: Demographic Information

Q1.A. What is your gender?

- ☐ Male
- ☐ Female
- ☐ Non-binary
- ☐ Prefer not to answer

Q1.B. What is your age?

- ☐ 18-22
- ☐ 23-25
- ☐ 26-34
- ☐ 35-54
- ☐ 55 or older

Q1.C. What is/are your native language/s?

- ☐ English

☐ German

☐ Other: Please Specify

Q1.D. How do you rate your level of English proficiency

☐ B2

☐ C1

☐ C2 or native

Q1.E. What is your current level of study?

☐ Bachelor

☐ Master

☐ PhD

☐ Other: Please Specify

Q1.F. What is your field of study (Studiengang)?

Q1.G. Please rate your level of personal experience with the following on a scale of 1 (very little experience) to 5 (very much experience):

	1 (very little experience)	2	3	4	5 (very much experience)
The Swiss legal system or Swiss law	☐	☐	☐	☐	☐
Data protection/privacy	☐	☐	☐	☐	☐
Programming	☐	☐	☐	☐	☐
Formal Logic	☐	☐	☐	☐	☐

Q1.H. Please select any of the following programming languages you have experience with and for the selected ones rate your proficiency in the following programming languages on a scale of 1 to 5, where 1 is beginner and 5 is expert:

	1 (beginner)	2	3	4	5 (expert)
R	☐	☐	☐	☐	☐
Python	☐	☐	☐	☐	☐
Java	☐	☐	☐	☐	☐
C/C++	☐	☐	☐	☐	☐
JavaScript	☐	☐	☐	☐	☐

	1 (beginner)	2	3	4	5 (expert)
Functional Programming (Haskell, Ocaml, etc)	☐	☐	☐	☐	☐
Prolog	☐	☐	☐	☐	☐
Others: Please Specify 	☐	☐	☐	☐	☐

. 2: Main Task

Q2.A. Please read the following Lex formalization of a legal provision regarding the validity of consent provided by users of an online platform.

article “1”

paragraph “1”

rule

whenever within 1d

IsFreelyGiven(u, c)

constitute

IsValid(u, c)

paragraph “2”

rule

whenever

IsMinor(u)

except

paragraph “1”

What legal text matches the rules defined by the following code?

- ☐ Article 1
1. *Consent shall be deemed valid if it has been freely given prior to the time when the consent's validity is assessed.*
 2. *Consent given by a minor shall under no circumstances be regarded as valid.*
- ☐ Article 1
1. *Consent shall be deemed valid if it has been freely given prior to the time when the consent's validity is assessed.*
 2. *Consent given by a minor shall not be regarded as freely given.*
- ☐ Article 1
1. *Users must give freely given and valid consent within one day of the time when the consent's validity is assessed.*
 2. *Consent given by a minor shall not be regarded as valid.*
- ☐ Article 1
1. *Consent shall be deemed valid if it has been freely given within one day prior to the time when the consent's validity is assessed.*
 2. *Consent given by a minor shall under no circumstances be regarded as valid.*

. 3: SUS survey

Q3.A. Please rate your agreement or disagreement with the following statements on a scale of 1 to 5, where 1 is strongly disagree and 5 is strongly agree

	1 (strongly disagree)	2	3	4	5 (strongly agree)
I think that I would like to use this language frequently	☐	☐	☐	☐	☐
I found the language unnecessarily complex	☐	☐	☐	☐	☐
I thought the language was easy to use	☐	☐	☐	☐	☐
I think that I would need the support of a technical person to use this language	☐	☐	☐	☐	☐
I found the various functions of the language well-integrated	☐	☐	☐	☐	☐
I thought there was too much inconsistency in this language	☐	☐	☐	☐	☐
I would imagine that most people would learn to use this language very quickly	☐	☐	☐	☐	☐
I found the language very cumbersome to understand	☐	☐	☐	☐	☐

	1 (strongly disagree)	2	3	4	5 (strongly agree)
I felt very confident understanding the language	☐	☐	☐	☐	☐
I needed to learn a lot of things before I could get going with this language	☐	☐	☐	☐	☐
I think that I would like to use this language frequently	☐	☐	☐	☐	☐

Block II

. 4: Reading Tasks

Q4.A. The following Lex code is an appropriate declaration for an event capturing: "a processor p processes data d as part of data processing task t ". You can assume that all processors are "entities"

```
event Process
```

```
  p: entity
```

```
  d: data
```

```
  t: data
```

- ☐ True
- ☐ False

Q4.B. The following Lex code is an appropriate declaration for an event capturing: "the legal basis of data processing task t is consent c ":

```
event HasConsentLegalBasis
```

```
t: task
```

```
c: consent
```

- ☐ True
- ☐ False

Q4.C. The following Lex code is an appropriate declaration for an event capturing: "controller c nominates processor p to perform data processing task t ":

```
event Nominate
```

```
c: entity
```

```
p: entity
```

- ☐ True
- ☐ False

Q4.D. What sentence describes best the rule defined by the following code?

rule

whenever

Process(p, d, t)

oblige

IsLegal(t)

- ☐ Any data processing task must be legal
- ☐ Data processing tasks are always legal
- ☐ Processing data is always legal
- ☐ Data can be processed legally only once

Q4.E. What sentence describes best the rule defined by the following code?

rule

whenever

IsFreelyGiven(co)

constitute

IsValid(co)

- ☐ When consent is freely given, it is considered valid
- ☐ When consent is freely given, it must be valid
- ☐ When consent is valid, it is considered freely given

Q4.F. What sentence describes best the rule defined by the following code?

rule

whenever

Process(p, d, t)

Relates(d, ds)

oblige

ONCE GiveConsent(ds)

- ☐ If some data is processed that relates to a data subject, then the data subject must have given consent at some time in the past or present.
- ☐ If some data is processed that relates to a data subject, then the data subject must give consent at some time in the present or future.
- ☐ If some data is processed, then that data must relate to a data subject and the data subject must have given consent at some time in the past or present.
- ☐ If some data is processed, then that data must relate to a data subject and the data subject must give consent at some time in the present or future.

. 5: Auditing Task

Q5. In this task, you will audit a Lex formalization of the following requirement from the General Data Protection Regulation (GDPR)

Article 7(1) GDPR

Where processing is based on consent, the controller shall be able to demonstrate that the data subject has

consented to processing of his or her personal data.

law GDPR

article 7

paragraph 1

law GDPR

type entity

type data

type data_subject

type activity

type consent

event Process

c: entity

a: activity

d: data

predicate PersonalData

d: data

ds: data_subject

predicate IsBasedOnConsent

a: activity

co: consent

event GiveConsent

ds: data_subject

co: consent

predicate IsAbleToDemonstrateConsent

c: entity

co: consent

rule

whenever

Process(c, a, d)

PersonalData(d, ds)

IsBasedOnConsent(a, co)

GiveConsent(ds, co)

oblige

IsAbleToDemonstrateConsent(c, co)

Reading

Controller **c** processes data **d** as part of activity **a**

Reading

Data **d** is personal data of data subject **ds**

Reading

Activity **a** is based on consent **co**

Reading

Data subject **ds** gives consent **co**

Reading

Controller **c** is able to demonstrate that consent **co** has been given

Reading

Whenever all of the following happen:

- Controller **c** processes data **d** as part of activity **a**
- Data **d** is personal data of data subject **ds**
- Activity **a** is based on consent **co**
- Data subject **ds** gives consent **co**

Then the following **shall be obligatory**:

- Controller **c** is able to demonstrate that consent **co** has been given

Where processing is based on consent, the controller shall be able to demonstrate that the data subject has consented to processing of his or her personal data.

. First, look at the above formalization on your own.

Q5.A. How would you rate your understanding of the formalization presented?

	1	2	3	4	5
1 = Very poor understanding 5 = Very good understanding	☐	☐	☐	☐	☐

Q5.B. Are there any specific aspects of the formalization that you find unclear or confusing? Please describe these aspects in detail.

Q5.C. Is the formalization in Figure 1 accurate compared to the text of Article 7(1) above? If not, what differences do you see between the formalization and the text? Please specify the discrepancies in detail.

. Now, pair up with your neighbor (lawyer + computer science pair) and take 5' to discuss the formalization. Then, answer the following questions.

Q5.D. How would you now rate your understanding of the formalization presented?

	1	2	3	4	5
1 = Very poor understanding 5 = Very good understanding	☐	☐	☐	☐	☐

Q5.E. Are there any specific aspects of the formalization that you still find unclear or confusing? Please describe these aspects in detail.

Q5.F. Is the formalization in Figure 1 accurate compared to the text of Article 7(1) above? If not, what differences do you see between the formalization and the text? Please specify the discrepancies in detail

Q5.G. If pairing up with your neighbor helped clarify your understanding, please explain how.

Block III

. 6: Post-Task Survey

Q6.A. What aspects of the language did you particularly like? [Please explain]

Q6.B. What aspects of the language did you find particularly difficult or confusing? [Please explain]

Q6.C. Do you have any suggestions for improving the language, especially in terms of usability?

Q6.D. Is there anything else you would like to share or comment on as part of this survey?